

Binomial probability in R

Mean and variance of a
Binomial

Normal approximation of a
Binomial

Linking to Pascal's triangle

Recap

L16: The Binomial Distribution Continued

Today's objectives

Binomial probability in R

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Recap

- ▶ Calculate exact and cumulative probabilities by hand and in R
- ▶ Find the mean and standard deviation of a Binomial distribution
- ▶ Decide when a Normal approximation is appropriate

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Recap

Remember: If X has the binomial distribution with n observations and probability p of success on each observation, the possible values of X are 0, 1, 2, ..., n . If k is any one of these values,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Binomial probability in R using `dbinom()`

- ▶ For discrete distributions we can calculate the probability of observing a specific value.
- ▶ For example, we can ask: What is the probability that exactly 3 of the ten bottles were contaminated when the risk of contamination was 10%?
- ▶ `dbinom()` is used to compute *exactly* 3

```
dbinom(x = 3, size = 10, prob = 0.1)
```

```
## [1] 0.05739563
```

Binomial probability in R using pbinom()

- ▶ Recall for Normal distributions we used `pnorm()` to calculate the probability *below* a given number.
- ▶ For our Binomial, we can also ask: what is the probability that 3 or fewer of the ten bottles were contaminated when the risk of contamination was 10%?
- ▶ `pbinom()` is used to compute 3 *or less*

```
dbinom(x = 3, size = 10, prob = 0.1)
```

```
## [1] 0.05739563
```

```
pbinom(q = 3, size = 10, prob = 0.1)
```

```
## [1] 0.9872048
```

Binomial probability in R

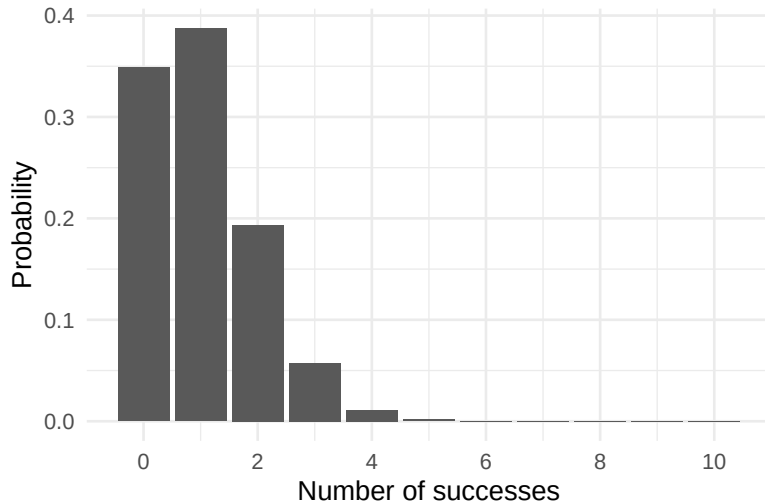
Mean and variance of a
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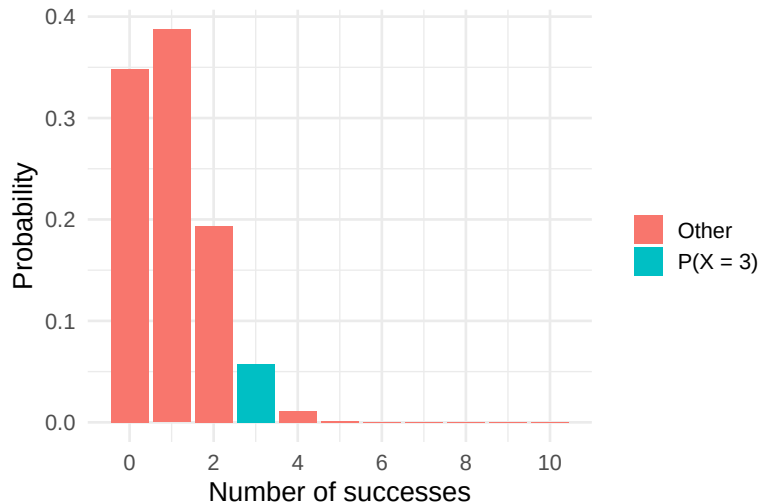
Recap

Histogram of binomial probabilities

This histogram shows the probability of observing each value of X . That is, it shows the $P(X = x)$, for x in $0, 1, 2, \dots, 10$, when $X \sim \text{Binom}(n = 10, p = 0.1)$

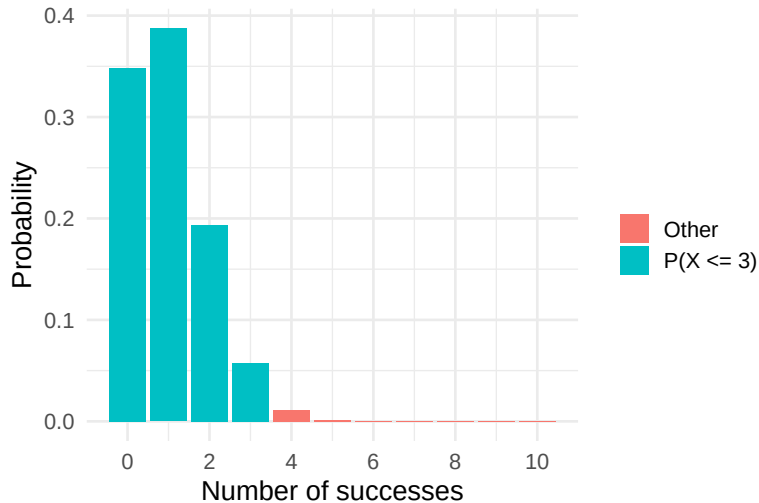


Exact discrete probability, graphed



```
dbinom(x = 3, size = 10, prob = 0.1)
```

Cumulative discrete probability, graphed



```
pbinom(q = 3, size = 10, prob = 0.1)
```

Your turn:

Imagine that someone installs a coffee machine in Berkeley Way West, but it is broken. 30% of the time, when a coffee is ordered it substitutes decaf for regular coffee. You are tasked with bringing coffees to your data project group meeting, and you order 4 coffees from the machine. What is the probability that everyone gets a caffeinated coffee? What is the probability that at least half of the group gets a caffeinated coffee?

Let X be the number of caffeinated coffees. Then $X \sim \text{Binom}(4, 0.70)$.

- ▶ Everyone: $P(X = 4) = 0.7^4 = 0.2401$
- ▶ At least half: $P(X \geq 2) = 1 - P(X \leq 1) = 0.9163$

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Mean and variance of a Binomial

Binomial mean and standard deviation

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Recap

If a count X has the binomial distribution with n number of observations and p as the probability of success, then the population mean and population standard deviation are:

$$\mu = np$$

$$\sigma = \sqrt{np(1 - p)}$$

Example of mean and SD calculations

Recall our example of the number of bottles contaminated in benzene, where $X \sim \text{Binom}(n = 10, p = 0.1)$.

$$\mu = np = 10 \times 0.1 = 1$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{10 \times 0.1(1-0.1)} = 0.9487$$

Thus, we **expect** to find one container contaminated with benzene per sample, on average. The standard deviation can be thought of, very roughly, as the expected deviation from this mean if you were to take many random samples.

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**Normal approximation of a
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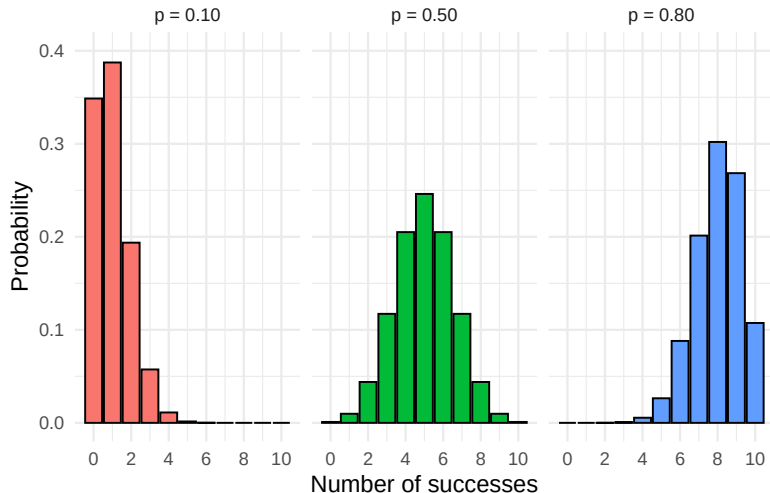
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Normal approximation of a Binomial

Histogram of binomial probabilities with different values for p

Here we have $n = 10$ and $p = 0.10, 0.50$, or 0.80 .



Histogram of binomial probabilities with different values for p

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Recap

How does the shape change when the probability is closer to .5?

What do you think happens when n gets larger?

An approximation to the binomial distribution when n is large

Imagine a setting where $X \sim \text{Binom}(n = 2000, p = 0.62)$. Then:

$$P(X = k) = \binom{2000}{k} 0.62^k (1 - 0.62)^{2000-k}$$

And:

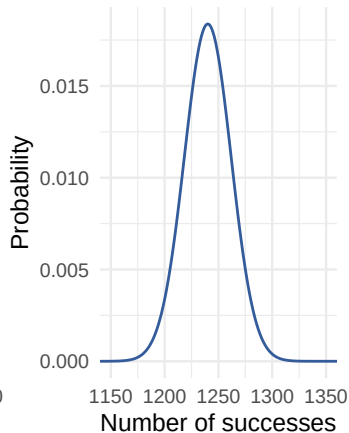
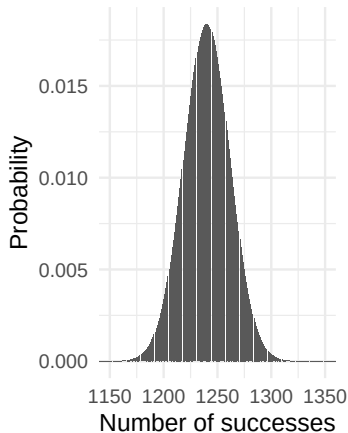
$$P(X \leq k) = \sum_{i=0}^k \binom{2000}{i} 0.62^i (1 - 0.62)^{2000-i}$$

If you were asked to calculate this by hand for, say, $k = 100$, it would take a very long time.

An approximation to the binomial distribution when n is large

Consider the probability distribution for $P(X = k) = \binom{2000}{k} 0.62^k (1 - 0.62)^{2000-k}$

What shape does this remind you of?



An approximation to the binomial distribution when n is large

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The previous graph is unimodal and symmetric. Let's calculate μ and σ :

$$\mu = np = 2000 \times 0.62 = 1240$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{2000 \times 0.62 \times (1-0.62)} = 21.70714$$

How much data is within 1 SD of the mean?

The interval 1240 ± 1 SD is $[1218.293, 1261.707]$.

Because X is integer-valued, the counts within this interval are 1219 through 1261. We can add their probabilities:

This code cycles through the probabilities to add them up

```
#students, no need to know how to write this code.
cumulative.prob <- 0

for(i in 1219:1261){
  cumulative.prob <- cumulative.prob + point.probs.2k[i + 1]
}

cumulative.prob

## [1] 0.6780514
```

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How much data is within 2 SD of the mean?

The interval $1240 \pm 2 \text{ SD}$ is $[1196.586, 1283.414]$.

Thus, we can use R to add up all the probabilities between $X = 1197$ and $X = 1283$ to give an approximate value for the probability within 2 SD of the mean:

This code cycles through the probabilities to add them up

```
#students, no need to know how to write this code.
cumulative.prob.2 <- 0

for(i in 1197:1283){
  cumulative.prob.2 <- cumulative.prob.2 + point.probs.2k[i + 1]
}

cumulative.prob.2

## [1] 0.9549547
```

► You could also perform the check for 3 SD

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The Normal approximation to Binomial distributions

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**Normal approximation of a
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Recap

From the previous calculations, you might see that the shape looks Normal and that the distribution nearly meets the 68%-95%-99.7% rule. Thus, it is approximately Normal.

This means that you can use the Normal distribution to perform calculations when data is binomially distributed with large n .

Example calculation of the Normal approximation to the Binomial

Suppose we want to calculate $P(X \geq 1250)$ using the Normal approximation. Without a continuity correction, we use 1250 as the cutoff:

```
# write the Normal code
```

```
1- pnorm(q = 1250, mean = 1240 , sd = 21.70714)
```

```
## [1] 0.3225149
```

Check how well the approximation worked:

```
# write the binomial code and see how well the approximation is
```

```
1 - pbinom(q = 1249, size = 2000, prob = 0.62)
```

```
## [1] 0.3313682
```

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Normal approximation for binomial distributions

Suppose that a count X has the binomial distribution with n observations and success probability p . When n is large, the distribution of X is approximately Normal. That is,

$$X \sim N(\mu = np, \sigma = \sqrt{np(1-p)})$$

As a general rule, we will use the Normal approximation when n is so large that $np \geq 10$ and $n(1-p) \geq 10$.

It is most accurate for p close to 0.5, and least accurate for p closer to 0 or 1.

Normal approximation with continuity correction

This approximation can be improved a tiny bit!

Counts can take only integer values, whereas a Normal variable is continuous. The count 1250 corresponds to the interval from 1249.5 to 1250.5. Therefore, $P(X \geq 1250)$ is approximated by the Normal area above 1249.5.

This correction makes a bigger difference when n is small.

```
1- pnorm(q = 1249.5, mean = 1240 , sd = 21.70714)
```

```
## [1] 0.3308222
```

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An aside: Pascal's triangle

$$\begin{array}{c} \binom{0}{0} \\ \binom{1}{0} \binom{1}{1} \\ \binom{2}{0} \binom{2}{1} \binom{2}{2} \\ \binom{3}{0} \binom{3}{1} \binom{3}{2} \binom{3}{3} \end{array}$$

Binomial probability in R

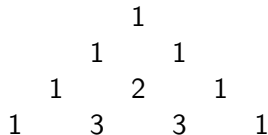
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An aside: Pascal's triangle



TED ed Video about Pascal's triangle

<https://www.youtube.com/watch?v=XMriWTvPXHI>

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Properties of the Normal distribution

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Recap

- ▶ the mean μ can be any value, positive or negative
- ▶ the standard deviation σ must be a positive number
- ▶ the mean is equal to the median (both $= \mu$)
- ▶ the standard deviation captures the spread of the distribution
- ▶ the area under the Normal distribution is equal to 1 (i.e., it is a density function)
- ▶ a Normal distribution is completely determined by its μ and σ

The 68-95-99.7 rule for all Normal distributions

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- ▶ Approximately 68% of the data fall within one standard deviation of the mean
- ▶ Approximately 95% of the data fall within two standard deviations of the mean
- ▶ Approximately 99.7% of the data fall within three standard deviations of the mean

Properties of the Binomial distribution

- ▶ A Bernoulli random variable must assume one of two possible and mutually exclusive outcomes
- ▶ Each trial of the BRV results in either a success or failure
- ▶ Each trial must be independent of every other trial
- ▶ Derived from the experiment: counting the number of occurrences of an event in n independent trials
- ▶ Random Variable: X = number of times the event happens in the fixed number of trials (n)
- ▶ Parameters
 - ▶ n = number of trials
 - ▶ p = probability of success (event happening)

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New R code:

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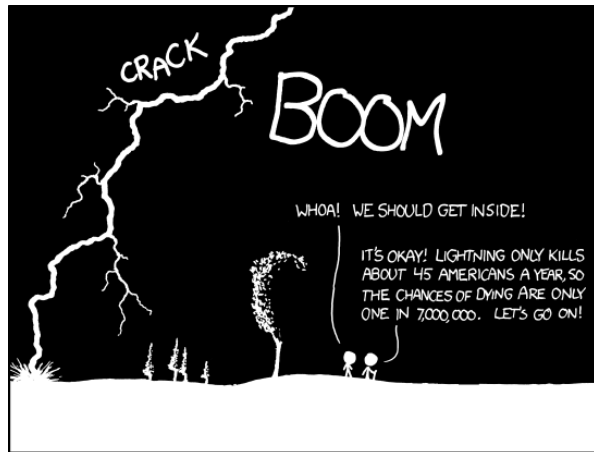
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Recap

- ▶ `pbinom(q,size, prob)` to calculate the cumulative probability under a binomial distribution

Comic Relief



THE ANNUAL DEATH RATE AMONG PEOPLE
WHO KNOW THAT STATISTIC IS ONE IN SIX.

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